

Seasonal Classification and Traffic Forecasting in Mobile Network Cells

Bachelor Project

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1 Introduction

Mobile networks are under constant pressure to deliver reliable service, while the resources required to do so, including frequency spectrum, infrastructure, and capacity, remain costly. When operators fail to anticipate future traffic demand, the consequences are wasted capacity, degraded user experience, and inefficient infrastructure investment. Beyond immediate performance, accurate traffic forecasting enables operators to make informed decisions about where and when to upgrade network infrastructure, ensuring that capital is directed toward cells where demand genuinely warrants it

Mobile network cells do not follow a single traffic pattern. Geographic location strongly influences both the level and timing of traffic. These differences are driven by human behavior, such as commuting and recreational activity, as well as network effects like load shifting between nearby cells. As a result, forecasting across all cells is a diverse time-series problem.

Because of this variation, building a robust forecasting model is not straightforward. One approach is to model each cell independently. This preserves local structure but is computationally expensive and can lead to overfitting, especially for cells with limited historical data. Another approach is to train a single global model across all cells. This allows shared representations of traffic patterns and can capture differences between cell types. However, it reduces interpretability and statistical consistency, since the model combines data from sites with different underlying traffic patterns. A third approach is to pre-classify cells based on their dominant traffic behavior, grouping cells with similar temporal structures. This reduces diversity within each group, making traffic patterns more consistent across cells within the same category. In this setting, simpler statistical models such as ARIMA or OLS can remain competitive when applied to more consistent subsets of cells.

This work investigates whether pre-classifying cells based on traffic behavior can improve forecasting performance compared to a single global approach. Using historical data from TDC NET’s mobile network, a data-driven classification method is developed to separate cells into seasonal and non-seasonal categories, and several forecasting models are evaluated within each group.

The remainder of this work first describes the dataset and preprocessing steps, then introduces a method for classifying cells based on seasonality and traffic behavior, and finally evaluates several forecasting approaches, including OLS, ARIMA, and XGBoost, in terms of forecasting performance.

2 Theory

2.1 Cellular Network Architecture

A cellular network is a wireless communication system that provides mobile devices with access to internet services and Mobile communication. To provide coverage over large geographical areas, network operators deploy base stations at different locations. Each

base station serves users within its coverage area and together they can cover wide geographical areas. As shown in Figure 1, multiple base stations can be deployed to form a connected network covering a wide area.

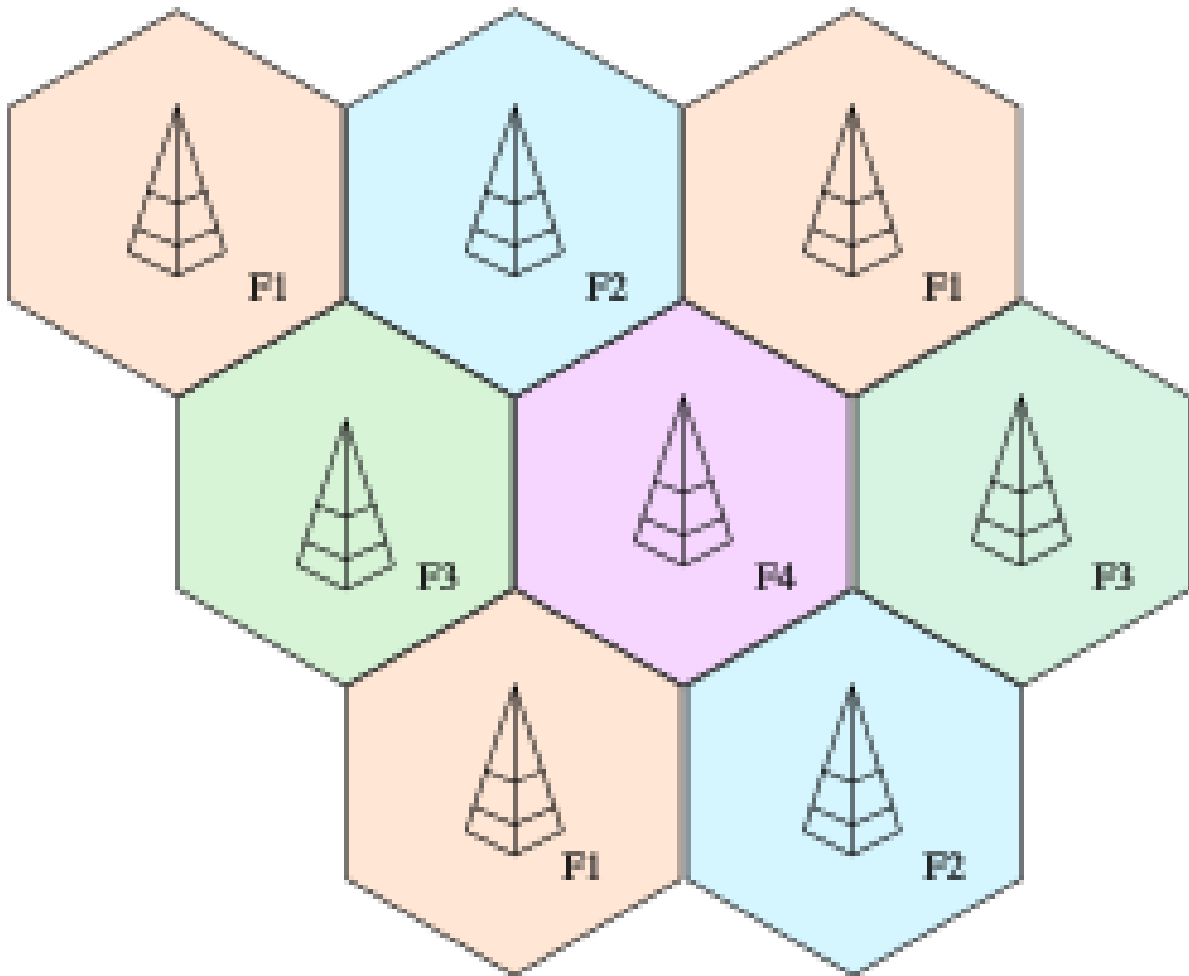


Figure 1: Cellular network architecture showing base stations, user devices, and inter-cell connectivity.

Modern base stations are commonly divided into sectors covering approximately 120 degrees each [1]. Within each sector, multiple cells may operate on different frequency bands and technologies, such as 4G and 5G. A base station consist of multiple cells and the cell is the basic unit of the network and handles communication between user devices and the network.

User devices connect to the network by attaching to the cell that provides the best radio conditions. This decision is mainly based on signal strength and signal quality, which are continuously measured by the device and the network. One key metric is the signal-to-interference-plus-noise ratio (SINR), which describes the quality of the received signal relative to background noise and interference from neighboring cells.

As users move or radio conditions change, connections may be transferred between cells in a process called handover [2]. Handover ensures continuous connectivity by moving

the user to a cell with better signal quality or more available capacity when needed.

2.2 Modulation and Signal Quality

Orthogonal Frequency Division Multiplexing (OFDM) is a multi carrier modulation technique that divides a wide frequency band into many narrow, parallel subcarriers [3]. Each subcarrier is shaped as a sinc function in the frequency domain, which produces a peak at its own center frequency and crosses zero at the center frequencies of all neighboring subcarriers as seen in figure 2.

This means that even though the subcarriers overlap in spectrum, sampling each one at its own center frequency yields no contribution from its neighbors, they are mathematically orthogonal. The subcarrier spacing is chosen carefully to be the inverse of the symbol duration, which guarantees these zero crossings align. This allows subcarriers to be packed far more tightly than traditional frequency division multiplexing, where guard bands would be required to prevent interference, resulting in much more efficient use of spectrum.

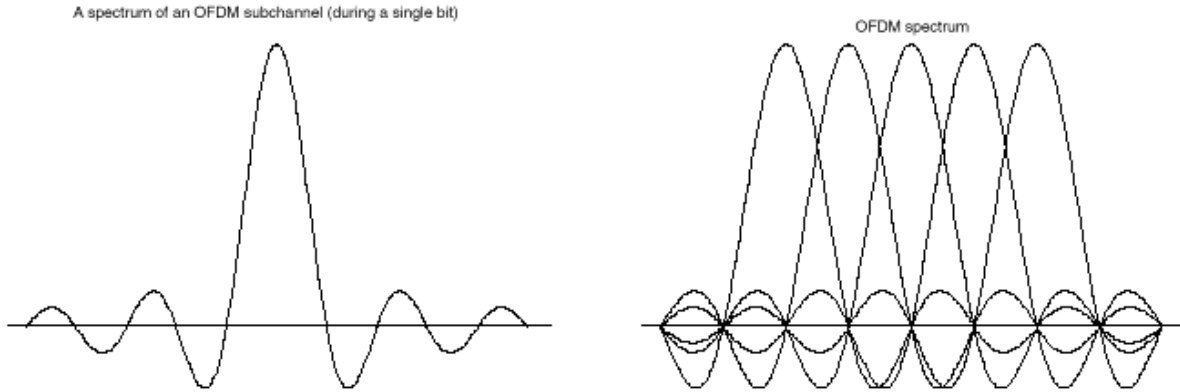


Figure 2: OFDM Sinc function alone vs with 5 subcarriers

Modern cellular networks employ adaptive modulation schemes, where the modulation order is selected according to the user's radio conditions [4]. Quadrature Amplitude Modulation (QAM) encodes information by varying both the amplitude and phase of the transmitted signal. Each combination of amplitude and phase corresponds to a unique symbol in the constellation diagram shown in Figure 3.

The number of bits carried by a symbol depends on the modulation order and can be expressed as

$$b = \log_2(M)$$

where M is the number of constellation points. Modulation schemes supported in LTE and NR systems include QPSK, 16QAM, 64QAM, and 256QAM in both uplink and downlink [4], corresponding to 2, 4, 6, and 8 bits per symbol respectively.

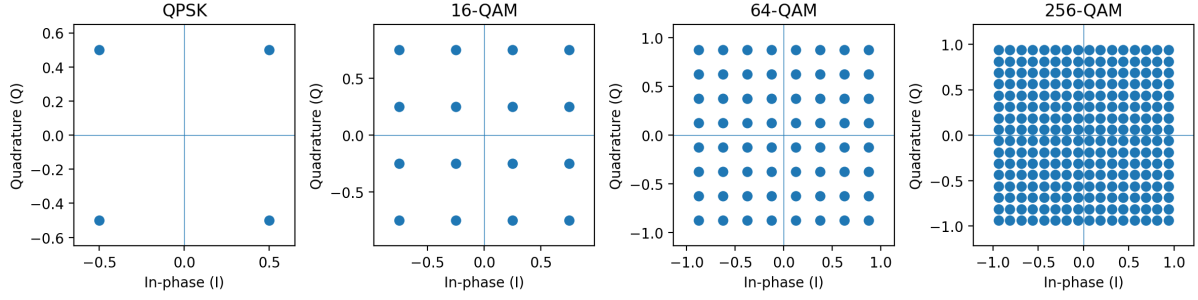


Figure 3: Constellation diagrams for different modulation schemes, illustrating the increasing number of symbols and bits per symbol from QPSK to higher-order QAM.

The modulation order selected by the scheduler depends on the radio conditions experienced by the user equipment, primarily expressed through the Signal-to-Interference-plus-Noise Ratio (SINR) or Channel Quality Indicator (CQI). The base station performs link adaptation based on these measurements, selecting an appropriate Modulation and Coding Scheme (MCS) to match the current channel quality [5]. As channel conditions improve, higher order modulation schemes can be used because constellation points can be placed closer together while remaining reliably distinguishable at the receiver. Conversely, under poor channel conditions, the system falls back to lower order modulation schemes, which are more robust to noise and interference but carry fewer bits per symbol.

The trade-off between modulation order and reliability can be quantified through the Bit Error Rate (BER). For square M-QAM, the theoretical BER as a function of the received E_b/N_0 is well approximated by:

$$\text{BER} \approx \frac{4}{\log_2 M} \left(1 - \frac{1}{\sqrt{M}} \right) \frac{1}{2} \text{erfc} \left(\sqrt{\frac{3 \log_2 M}{2(M-1)}} \cdot \frac{E_b}{N_0} \right)$$

where E_b/N_0 is the received energy per bit to noise ratio. As shown in Figure 4, higher-order modulation schemes require a significantly larger E_b/N_0 to achieve the same bit error rate, reflecting the reduced distance between constellation points in figure 3

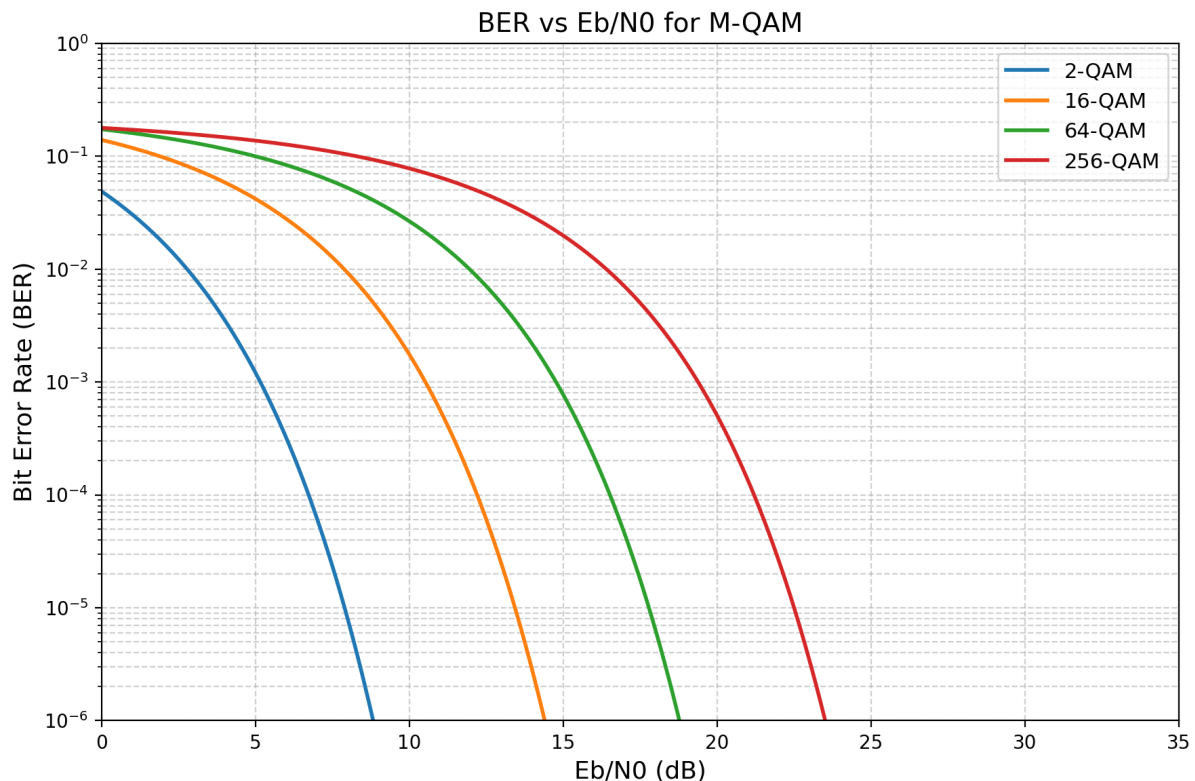


Figure 4: Bit Error Rate plot of different QAM

The result is that two cells with identical PRB utilization can deliver very different data volumes depending on the network conditions experienced by their users, making SINR a fundamental determinant of cell capacity. Beyond its role in per user throughput, SINR is also an important indicator in network planning. Areas with consistently low SINR suggest poor coverage, which may indicate the need for additional infrastructure such as new base stations or upgrades to existing cells. In this way, signal quality metrics inform not only real-time scheduling decisions but also longer-term capacity planning.

2.3 Resource Allocation and PRB Utilization

In 5G NR and 4G LTE, the available frequency spectrum is organized into a two dimensional time frequency grid as seen in figure 5. The frequency axis is 180kHz wide divided into subcarriers spaced 15kHz apart [4], while the time axis is divided into symbols. A Physical Resource Block (PRB) consists of 12 consecutive subcarriers in the frequency domain [6]. This grid structure is a direct consequence of the OFDM waveform described in the previous section, where each subcarrier occupies a defined 15kHz position in the frequency domain, without any cross talk.

Each cell allocates its available PRBs between connected users through a scheduling process that runs every millisecond. During each scheduling interval, the base station evaluates all active users, their current data demand, and their radio conditions. Based on this information, PRBs are assigned to users according to a scheduling algorithm that balances throughput, fairness, and service requirements.

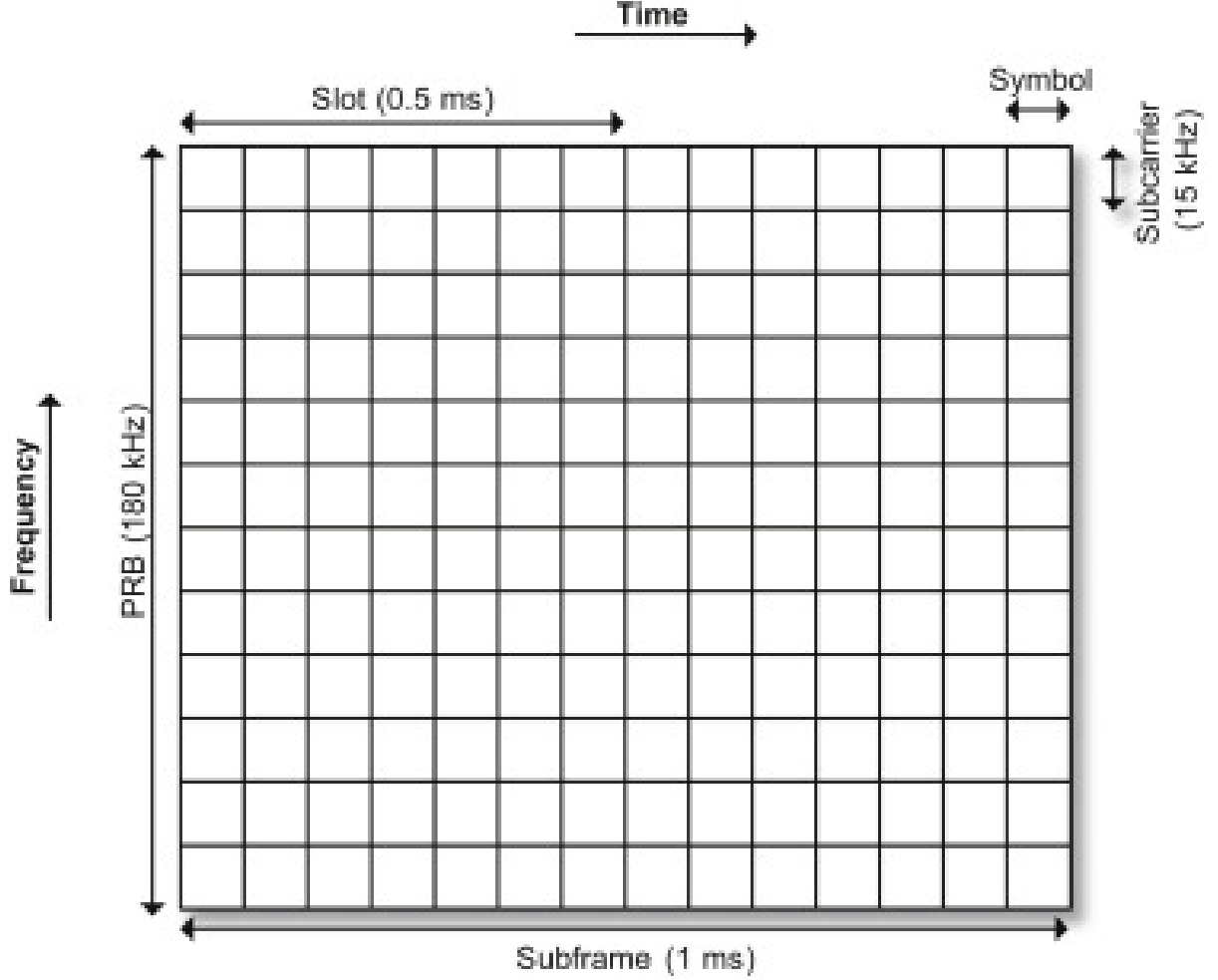


Figure 5: PRB Time and Frequency resource grid

Proportional Fair scheduling

The most commonly used approach is Proportional Fair scheduling algorithm, it aims to achieve a trade off between maximizing total throughput and ensuring fairness among users with different radio conditions [7]. The key idea is to prioritize users who currently have good channel conditions relative to their past achieved throughput.

The PF scheduler computes a scheduling metric for each active user at scheduling time t :

$$M_i(t) = \frac{T_{\text{inst},i}(t)}{[T_{\text{average},i}(t)]^a}$$

where $T_{\text{inst},i}(t)$ is the instantaneous achievable throughput derived from channel quality indicator and the corresponding modulation and coding scheme, and $T_{\text{average},i}(t)$ is the long-term average throughput for user i , obtained using an exponentially weighted low-pass filter. The parameter a controls the fairness-throughput trade-off and is typically set to 1 [7].

Each interval is divided into N time slots, as shown in Figure 5. These time slots are

assigned sequentially. After each assignment, the PF metric and throughput averages are updated before the next partition is scheduled. This allows multiple users to be served within a single interval, while still allowing a single user to receive multiple time slots within the same interval depending on their metric.

At each scheduling step, the user with the highest metric is selected for allocation of the current resource unit. This process is repeated over all fixed scheduling resources within the interval rather than resulting in a single user selection for the entire interval.

PRB Utilization

PRB utilization measures the fraction of available resource blocks that are actively used during a reporting interval. It reflects the overall load of the cell by indicating how much of the available capacity is occupied at a given time, independent of how efficiently individual resource blocks are utilised.

High PRB utilization indicates a heavily loaded cell where most available resources are in use, while low utilization indicates available capacity. However, PRB utilization alone does not fully characterize cell performance, since it does not account for the efficiency with which each block is used. The relationship between PRB utilization and throughput provides a more complete picture of cell operating conditions, as illustrated in Figure 6.

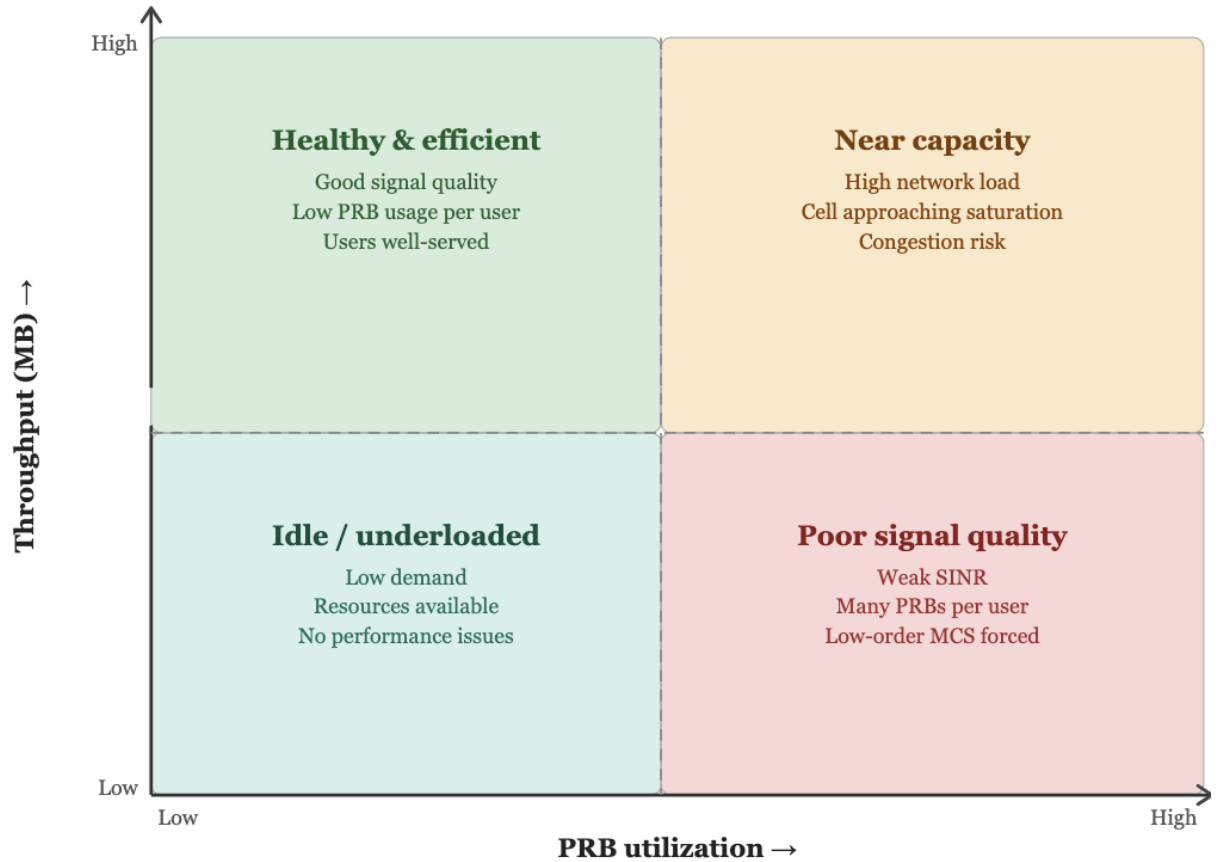


Figure 6: PRB utilization vs Throughput

Four general operating states can be identified from the relationship between throughput and PRB utilization as shown in figure 6. Low PRB utilization combined with low throughput indicates an underloaded cell, where little traffic is being served and most resources remain idle. High PRB utilization combined with low throughput indicates inefficient resource use, typically caused by poor radio conditions that force the scheduler to allocate more PRBs per unit of data. High throughput combined with low PRB utilization indicates efficient operation under favorable radio conditions, where higher-order modulation allows more data to be carried by fewer blocks. Finally, high PRB utilization combined with high throughput indicates a heavily loaded cell operating close to its capacity limit, where resources are being used efficiently but congestion risk increases during periods of peak demand.

Among these states, high PRB utilization combined with low throughput represents the least efficient operating condition. It indicates that the cell is consuming a large share of its available radio resources while delivering relatively little user data, typically as a consequence of degraded signal quality across many connected users.

2.4 Uplink and Downlink Traffic

Traffic in cellular networks is directional. Downlink refers to data transmitted from the network to the user device, while uplink refers to data transmitted from the user device to the network. Applications such as video streaming, web browsing, and social media primarily generate downlink traffic, whereas uplink traffic typically consists of user-generated content, control signalling, and other lower-volume data transmissions. As a result, downlink traffic generally dominates in consumer mobile networks.

This asymmetry is reflected in how cellular systems are designed. In Time Division Duplex (TDD) systems, uplink and downlink share the same frequency band but are separated in time [4]. The available resources are divided into time slots that are assigned to either uplink or downlink, allowing the network to adapt to traffic demand. In contrast, Frequency Division Duplex (FDD) systems use separate frequency bands for uplink and downlink, enabling simultaneous transmission in both directions.

In both cases, the overall resource allocation is typically biased heavily toward downlink traffic to reflect the dominant direction of data usage in mobile networks.

2.5 Traffic Generation in Cellular Networks

Cellular network traffic originates from individual user sessions, where a session refers to any period of data exchange between a user's device and the network cell. These sessions can vary widely in terms of duration, data volume, uplink/downlink depending on the application. Streaming videos generate a sustained high volume downlink session, while messaging generates short bursts of low volume traffic, and audio calls generate a sustained low volume traffic.

At any given time, the traffic observed at the cell level is the sum of all active user sessions

within that cell. This means that cell level traffic reflects the combined behavior of many independent users rather than a single transmission pattern. As a result, traffic at the network level follows human activity patterns. Users do not generate sessions uniformly over time, but instead concentrate activity around daily routines such as working hours and commuting. This leads to clear daily patterns, with high load during the day and lower activity during the night. Longer-term variations can also appear, including weekly differences between weekdays and weekends, as well as seasonal effects driven by travel and population movement.

Different geographical environments produce structurally distinct traffic patterns. Cells located near event venues such as stadiums and exhibition centers may experience long periods of low activity interrupted by short, sharp peaks triggered by external events. These series are non-stationary, since their statistical properties change abruptly during event windows.

Recreational cells serving destinations such as ski resorts, coastal regions, and holiday areas follow smoother and more regular seasonal cycles. These patterns are visible across months or years, with recurring peaks during specific periods. Traffic in these cells is typically not stable week to week, but exhibits seasonal cycles that repeat year to year.

Urban and commercial cells tend to exhibit a more stable traffic structure. Traffic is driven by regular human activity patterns such as work commutes and daily routines, producing strong and consistent daily and weekly cycles, including differences between weekdays and weekends. These series tend to be closer to stationarity compared to recreational or event-driven cells.

This user activity forms the basis of the measurements collected by the network. The base station continuously records and summarizes these events into cell level indicators, which are then reported as Performance Management (PM) counters at fixed time intervals.

2.6 PM Counters

Mobile network operators typically collect standardized measurements from their network cells using Performance Management (PM) counters. These are software counters embedded in the base station that continuously record network activity and report values at fixed time intervals, typically every 15 minutes or every hour, depending on the network configuration.

PM counters operate at the cell level, meaning that each counter value represents the combined activity of all users connected to a specific cell during the reporting interval. Typical counter categories include traffic volume counters, PRB utilization counters, unique user counters, and signal-to-noise ratio counters. Traffic measurements include both downlink and uplink counters, and most categories are reported as average and maximum values over the interval.

Traffic volume counters are typically implemented as cumulative sums of transmitted

and received data bytes at the base station. These counters are incremented based on the actual number of successfully transmitted bits at the MAC or PDCP layer during scheduling. At the end of each reporting interval, the accumulated values are exported and reset or reported as deltas, and higher-level KPIs such as throughput are derived from these measurements.

3 Dataset and Preprocessing

The dataset used in this study is sourced from TDC NET as a part of a collaboration in connection with this bachelor’s thesis. It consists of hourly time-series measurements collected from 12 cellular network towers across Denmark. The towers are grouped into three categories, urban, recreational, and suburban. The labels are based solely on the geographical location of each tower. Overall summary of the dataset is shown in table 1.

Table 1: Dataset before preprocessing

Statistic	After
Rows	3,104,976
Urban sites (cells)	4 (83)
Suburban sites (cells)	4 (93)
Recreational sites (cells)	4 (71)
Cells total	255
Missing rate (data volume dl)	33.2%
Start date	2024-05-21
End date	2026-03-01

The dataset is structured as a multivariate time series. Each row corresponds to an hourly observation, while each column represents a performance indicator describing the network traffic during that hour. Several features are further separated by traffic direction (uplink and downlink). An overview of the features is provided in Table 2.

To provide a clearer visualization of the dataset, Figure 7 shows the daily peak downlink data volume across all cells for two example towers, a recreational tower (R2) and an urban tower (U2). The figure also reveals a gap in August 2024 due to missing data in the dataset.

Already at this stage, clear differences in traffic behavior can be observed between the two tower types. The urban tower exhibits a relatively stable and gradually increasing trend over time, which suggests linear trend behavior. In contrast, the recreational tower displays strong seasonal variations, with traffic volumes increasing significantly during the summer months. These differences illustrate the differences in traffic volume across tower types and underscore the importance of treating each category separately in subsequent analysis.

Feature	Type
Signal-to-noise ratio (SINR max)	continuous
Signal-to-noise ratio (SINR avg)	continuous
Peak average users	continuous
Average users	continuous
Peak data volume Downlink	continuous
Peak data volume Uplink	continuous
Average data volume Downlink	continuous
Average data volume Uplink	continuous
PRB utilization Downlink	continuous
PRB utilization Uplink	continuous
Average user throughput Downlink	continuous
Average user throughput Uplink	continuous
Site ID	categorical identifier
Cell ID	categorical identifier
Frequency technology (4G/5G)	categorical
Frequency spectrum	categorical

Table 2: Overview of dataset features and their types

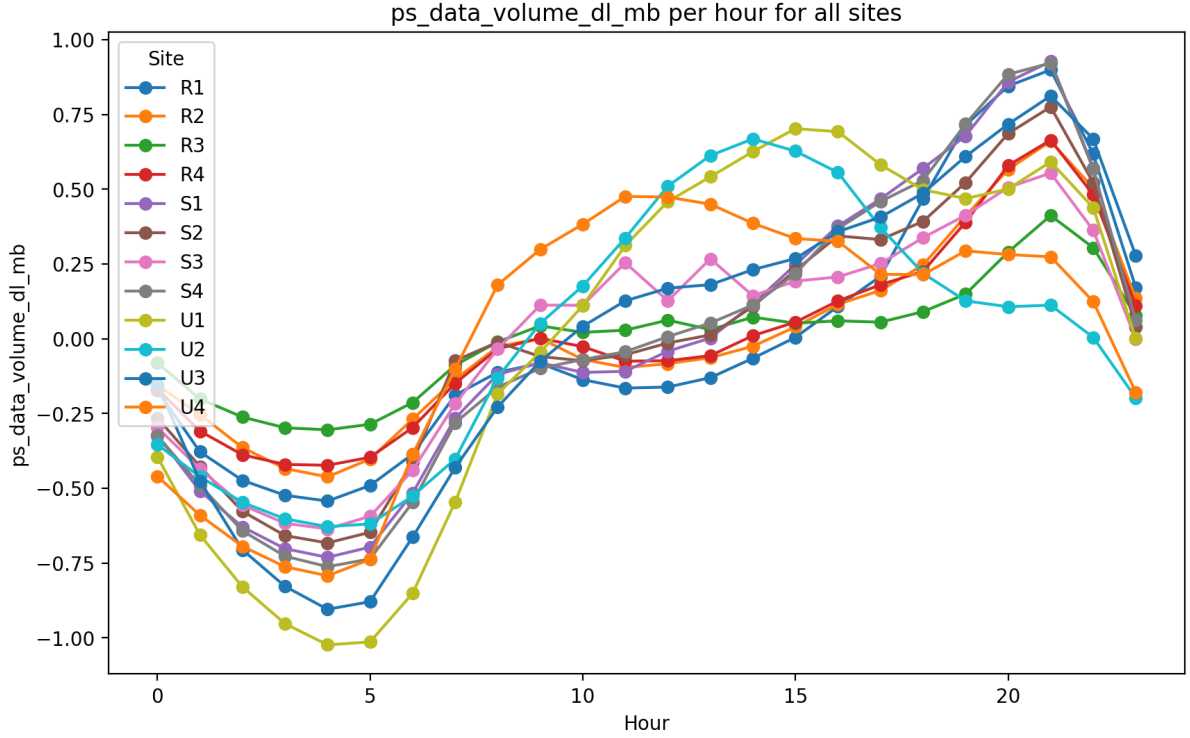


Figure 8: Average traffic volume per hour of day across all cells

Figure 8 shows the average hourly downlink data volume for each site, normalized using z-score standardization, to get a clear comparison in patterns across the sites. Clear differences can be seen between urban, suburban, and recreational cells.

All site types reach their minimum traffic between 03:00 and 05:00, with urban cells (U1–U4) showing the lowest values, with z-scores around (-0.75 to -1). This suggests that

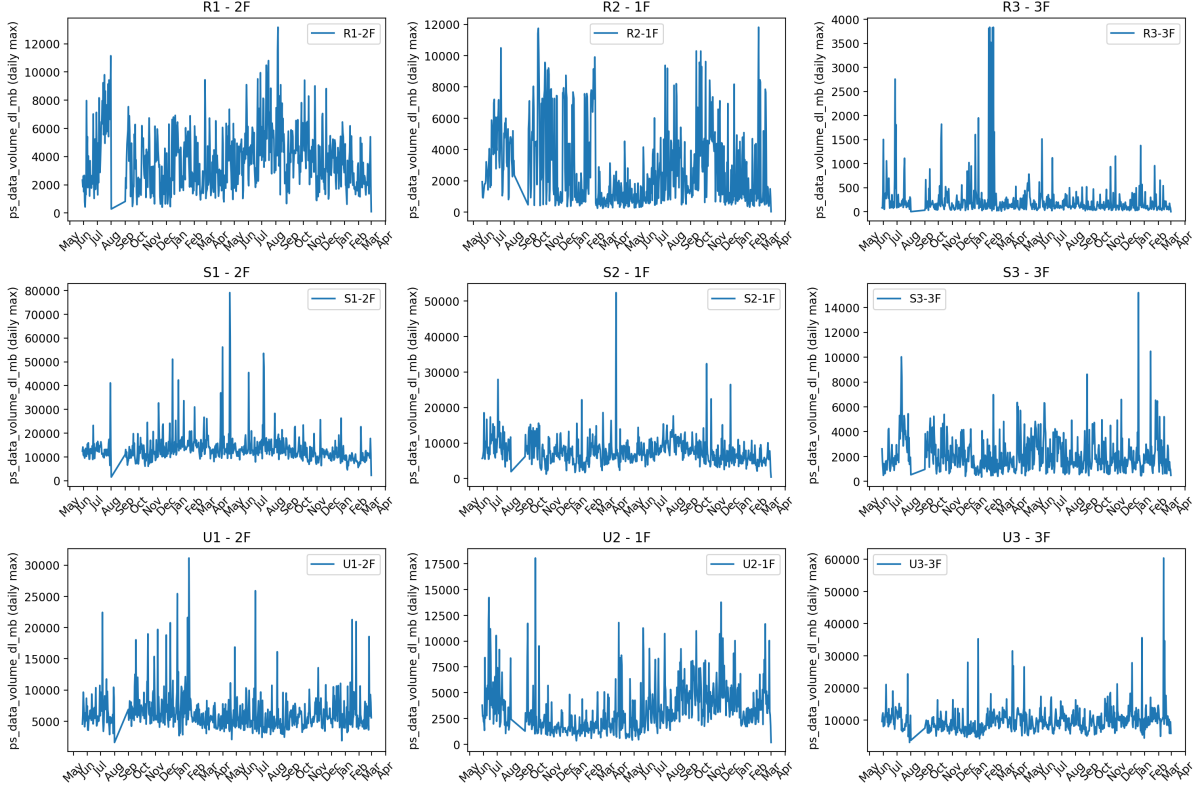


Figure 7: 2F cell Daily peak data volume downlink across all sites

urban cells experience the strongest night-time drop relative to their average level, reflecting a clear difference between peak and off-peak demand in densely populated areas..

Urban cells recover quickly from the overnight minimum, reaching a mid day peaking around 11:00-16:00 at z-scores of approximately 0.5-0.65, and declining again after 16. By contrast, suburban and recreational cells remain relatively flat during midday, hovering near their mean ($z\text{-score} \approx -0.1 - 0.1$) throughout the middle of the day.

The most striking divergence occurs in the evening. Recreational and suburban cells reach their daily peak between 19:00 and 21:00, with several sites such as R1, S4, and U3 exceeding a z-score of 0.75. Urban cells also rise in the evening but peak slightly earlier, around 20:00, and at comparatively lower normalized levels. This represents an evening peak roughly 30–40% higher in normalized terms than the midday plateau observed in urban cells, highlighting a fundamentally different daily usage cycle.

Notably, U3 and R2 deviate from the dominant patterns within their respective categories. U3 follows a pattern more similar to recreational sites, with low activity during the day and a pronounced evening peak around 20:00. In contrast, R2 shows higher activity earlier in the day, reaching its highest values around 11:00 before gradually declining throughout the afternoon and evening.

These results confirm that the timing of peak demand differs systematically by cell type. Urban cells follow a midday-dominated pattern consistent with workplace and commuter activity, while recreational and suburban cells follow an evening-dominated pattern more

consistent with leisure usage. This distinction has direct implications for forecasting, as models trained on mixed cell types risk combining two structurally different daily cycles.

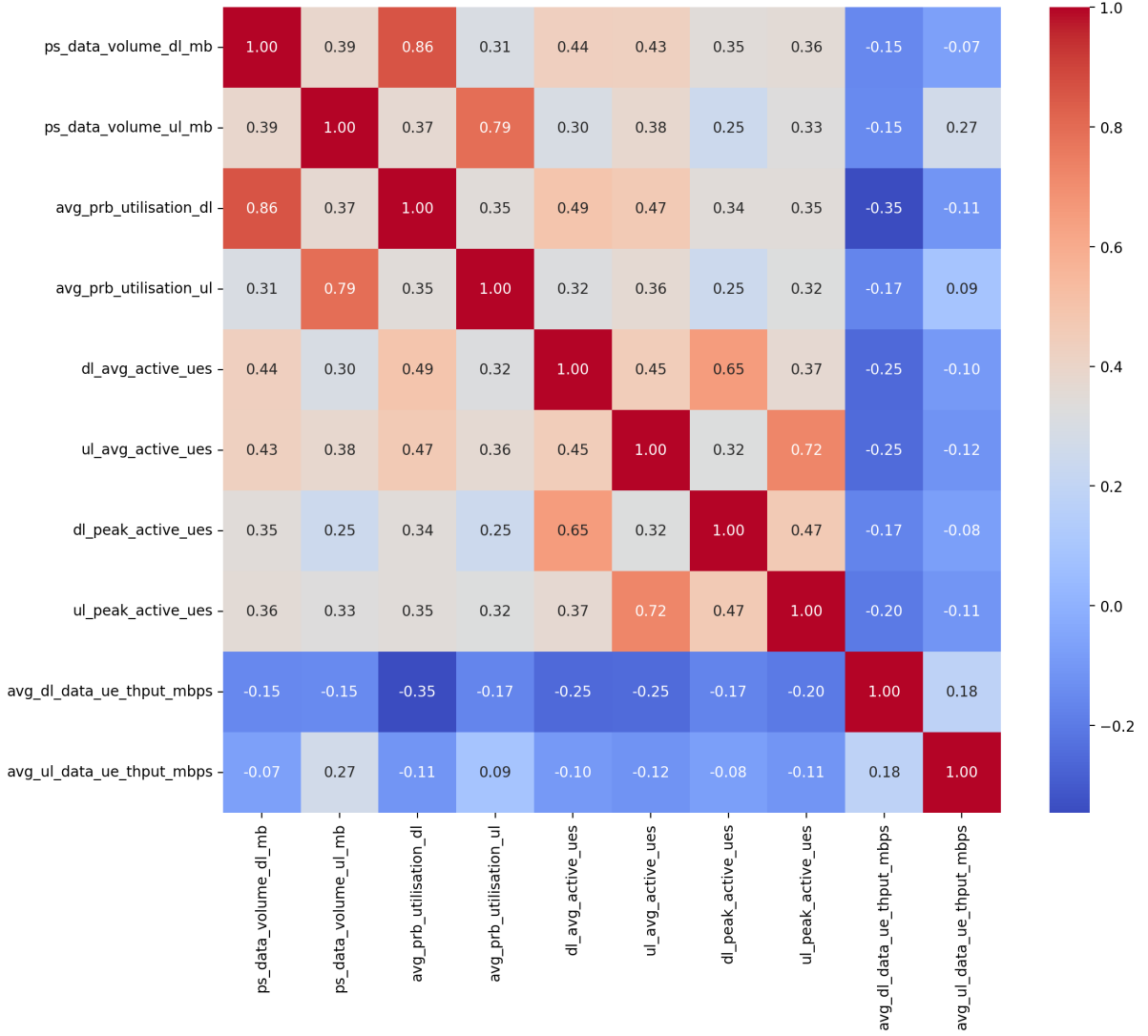


Figure 9: Feature correlation matrix

Figure 9 shows the correlation matrix of all continuous features in the dataset. Most features are positively correlated with each other. The main exception is user throughput (Mbps) negative correlations compared to the other features.

This behavior is expected. When a cell has fewer active users, it can allocate more throughput per user, which can lead to higher throughput values per user when overall traffic is low.

In general, features within the same category show strong correlations, such as average and peak values of the same feature. Another strongly correlated feature is the relationship between PS data volume and average PRB utilization, where higher traffic volume leads to higher resource usage in the cell.

Overall, the results suggest that traffic related metrics are strongly interdependent, meaning that a few key features, such as data volume, already provide a good representation of overall cell.

3.1 Preprocessing

Preprocessing is a crucial step in the analysis, particularly when the goal is to capture seasonal patterns in the data. To reliably observe seasonality, each time series must cover at least one full yearly cycle. Shorter time series may not contain enough information to capture these effects.

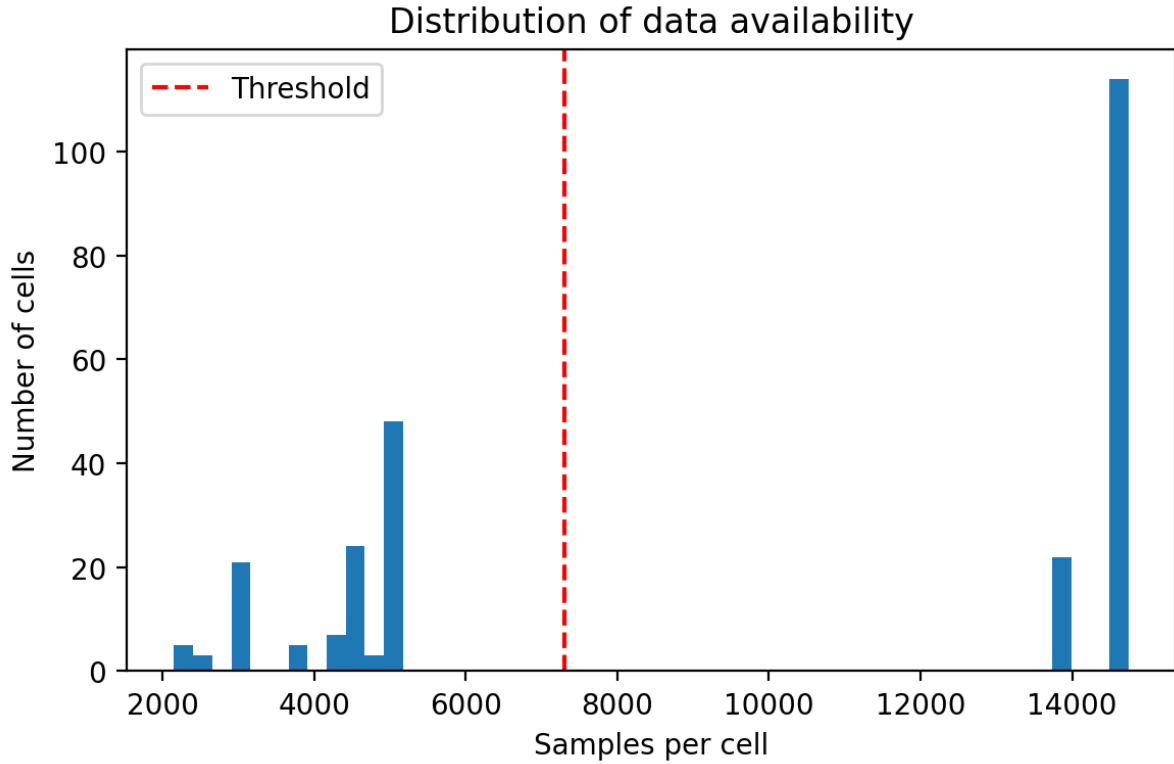


Figure 10: Distribution of available data samples per cell

Figure 10 shows the distribution of the number of observations per cell. Two distinct groups can be observed. The first group contains cells with approximately 15.000 samples, corresponding to more than one year of data. The second group contains cells with roughly 2000-5000 samples, which is below one year of observations.

This resulted in the removal of 125 cells due to insufficient length. The removals were distributed across site types as follows: Urban (44), Suburban (48), Recreational (33).

As we saw in figure 7, the month of august in 2024 is missing, this months contained 166 samples across all cells, which is way too little and was therefor removed.

Duplicate entries at the (site, cell, datetime) level were removed. When duplicates contained conflicting values, rows with non-missing values in the ps data volume dl column

were prioritized. This resulted in a deletion of 1,130,288 rows and the dataset going from 3,104,976 to 1,974,688 rows.

Table 3: Dataset after preprocessing

Statistic	After
Rows	1,770,806
Urban sites (cells)	4 (39)
Suburban sites (cells)	4 (45)
Recreational sites (cells)	4 (38)
Cells total	122
Missing rate (data volume dl)	166 (0.01%)
Start date	2024-05-21
End date	2026-03-01

Z-score normalization

Since the analysis focuses on patterns across cells rather than absolute values, Z-score normalization is applied to each cell individually. This transformation ensures that each time series has a mean of 0 and a standard deviation of 1.

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Here, μ is the mean and σ is the standard deviation of the values within each cell.

By applying this normalization, differences in scale between cells are removed, allowing for direct comparison of patterns in traffic load rather than magnitude differences.

4 Network Site Classification

Several approaches can be used for classifying time series data. Unsupervised learning methods such as k-means clustering are commonly used when the structure of the data is unknown. These methods can capture complex patterns but are often less interpretable. Alternatively, statistical approaches based on hypothesis testing provide more transparent decision rules but may rely on stronger assumptions about the data.

In this study, the objective is to separate cells into two classes, seasonal and non-seasonal. Given the binary nature of the problem and the need for interpretability, a simple and explainable approach is preferred over more complex clustering methods.

To identify a simple and interpretable seasonal indicator, a binary feature based on public school holiday schedules in Copenhagen [reference] was constructed. Each time series was labeled as either occurring during a holiday period or a non-holiday period.

For each cell, the traffic data was divided into two groups corresponding to holiday and non-holiday periods. The average traffic volume was then computed for both groups. To obtain an overall overview at the site level, the average values across all cells within each site were visualized using boxplots, as shown in Figure 11.

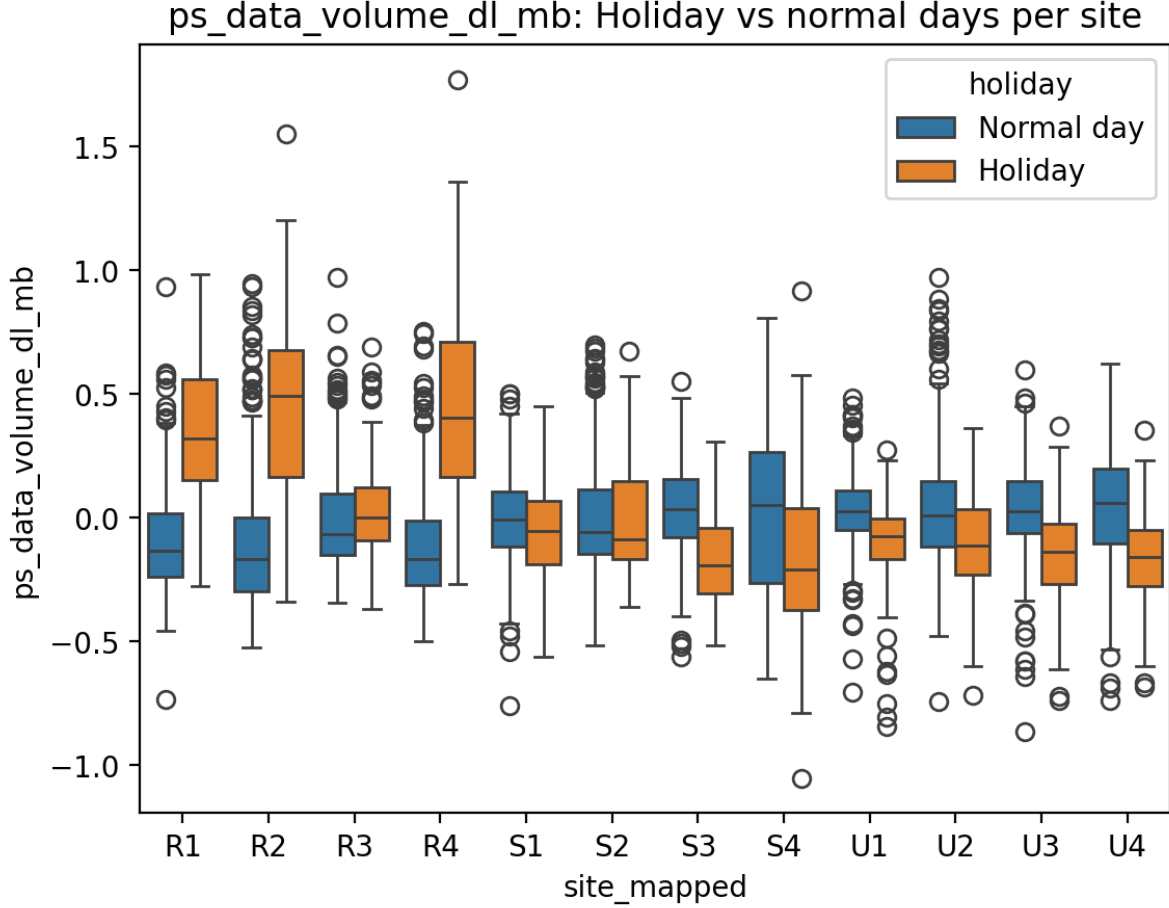


Figure 11: T-test holiday volume vs non-holiday volume for all individual cells

Figure 11 shows differences in standardized traffic levels between holiday and non-holiday periods across site categories.

Recreational sites (R1–R4) show a consistent positive shift in standardized traffic during holiday periods. The mean difference between holiday and non-holiday periods ranges from 0.04 to 0.58 standard deviations. The strongest effects are observed at R4 ($\Delta = 0.582$) and R2 ($\Delta = 0.547$), followed by R1 ($\Delta = 0.442$). R3 shows only a weak shift ($\Delta = 0.051$), indicating a much smaller seasonal response compared to the other recreational sites.

Urban sites (U1–U4) show a consistent negative shift during holiday periods. Mean differences range from -0.13 to -0.20 standard deviations, with the strongest decreases observed at U4 ($\Delta = -0.202$) and U2 ($\Delta = -0.192$). This indicates lower standardized traffic levels during holidays compared to normal days across all urban locations.

Suburban sites (S1–S4) show mixed and weaker effects. S1 ($\Delta = -0.039$) and S2

($\Delta = -0.020$) show near-zero changes, indicating little difference between holiday and non-holiday periods. S3 ($\Delta = -0.213$) and S4 ($\Delta = -0.141$) show larger negative shifts, more similar to urban patterns. Overall, the suburban group shows higher diversity, with no consistent directional response across all sites.

Median differences follow a similar pattern, with recreational sites showing positive shifts and urban sites showing negative shifts, confirming that the observed effects are not driven solely by extreme values.

4.1 T-test Validation

The boxplot analysis demonstrates that holiday periods produce different traffic volumes depending on site category, suggesting that the holiday indicator can serve as a basis for statistical classification at the cell level. To formalize this, an independent two-sample t-test is applied to each cell individually, testing whether the mean traffic volume during holiday periods differs significantly from that during non-holiday periods.

The t-statistic is defined as:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (2)$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, s_1^2 and s_2^2 are the sample variances, and n_1 and n_2 are the number of observations in the holiday and non-holiday groups.

The t-test is well suited to this problem for several reasons. First, the two groups, holiday and non-holiday periods, are naturally defined and non-overlapping. Second, the test provides a statistical measure of whether the difference in mean traffic between the two groups is likely to occur by chance. Based on this, a simple decision rule is introduced, cells with statistically significant differences in mean traffic ($p < 0.05$) are labeled as seasonal, while the remaining cells are labeled as non-seasonal. This produces an interpretable and reproducible classification criterion without requiring manual threshold tuning.

Figure 12 shows the t-test results across all individual cells. A clear separation is observed, with the majority of cells belonging to recreational sites identified as seasonal, while urban cells are predominantly identified as non-seasonal. This alignment with the geographic site categories suggests that the method captures meaningful differences in traffic behavior without using geographic location as an input.

The suburban sites present a more mixed case, as their traffic behavior does not conform as clearly to either category. Cells from suburban sites are distributed across both classes, reflecting the mixed nature of these environments. Because the method produces consistent results for both recreational and urban sites, it provides a useful basis for analysing suburban cells, where no clear ground truth classification exists. This highlights the value of a data driven approach over simple geographic labeling, as it can distinguish cases where location alone is not enough to determine traffic behavior.

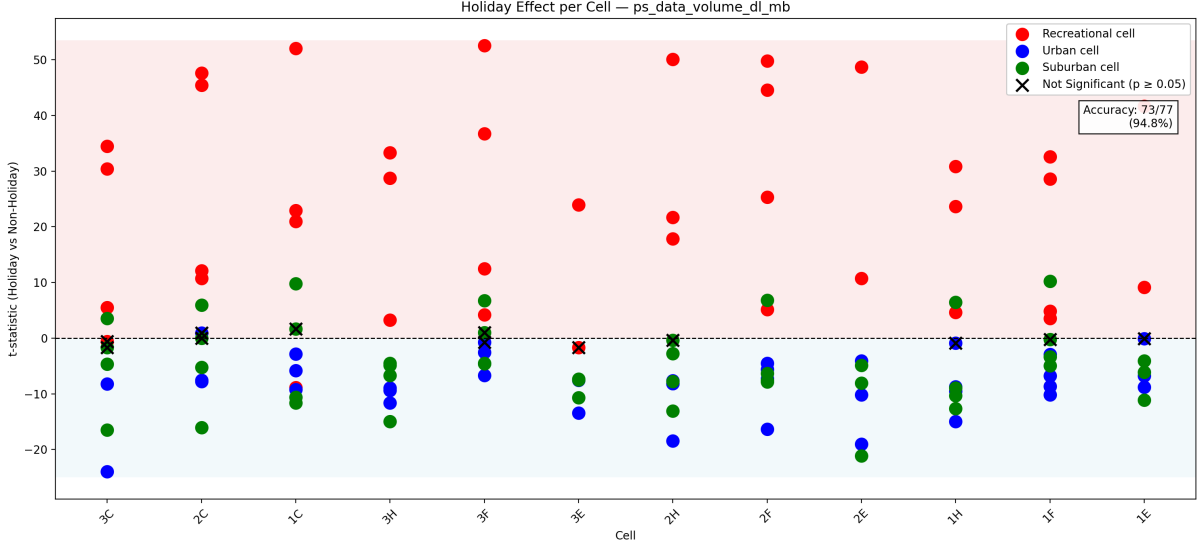


Figure 12: T-test holiday volume vs non-holiday volume for all individual cells

5 Traffic Forecasting

This section builds on the classification results presented earlier. The motivation for classification is to better understand underlying traffic patterns before performing forecasting. By separating network sites into seasonal and non-seasonal groups, it becomes possible to evaluate whether different traffic dynamics require different forecasting strategies.

Mean Squared Error

Doing this analysis the different models will be evaluated using the Mean Squared Error (MSE) which is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value. MSE penalizes larger errors more heavily due to the squaring term, making it particularly sensitive to large deviations, which is desirable in a network planning context where significant underestimation of peak traffic can lead to network overload.

5.1 Baseline Models

Before introducing more complex forecasting approaches, several baseline models are defined to provide reference performance levels. These baselines allow for a fair evaluation of whether more advanced methods provide meaningful improvements.

The dataset is evaluated under three settings, seasonal cells, non-seasonal cells, and a combined dataset. This enables analysis of whether traffic behavior differs across network environments. The cells were split into the different groups by the t-test as shown in figure 2, if the p-value was below 0.05, there was not enough evidence that the cell was

seasonal and it was put in the non-seasonal category.

The forecasting target is the daily peak traffic volume, defined as the maximum observed hourly traffic within a day as seen in figure 7. This metric is particularly important for network planning, as it reflects peak load conditions that are critical for capacity management.

Model	All	Recreational	Urban
Seasonal Naive	7.57	7.96	7.37
Mean Naive	4.62	3.98	4.91
Previous Naive	3.56	3.57	5.01

Table 4: Mean Squared Error (MSE) for baseline forecasting models across different site categories

Table 4 presents the baseline model results evaluated on the final two months of the dataset (January–March 2026). The seasonal naive model, which forecasts each period by repeating the corresponding values from the previous year, performs worst across all splits. This suggests that traffic patterns are not stable year over year.

The mean naive model forecasts using the training set mean, while the previous naive model uses the most recently observed value as the prediction. A notable pattern is seen when comparing the two across dataset splits, the mean model performs better on non-seasonal cells, while the previous value model performs better on seasonal cells. This is consistent with the underlying traffic patterns, non-seasonal cells shows relatively stable behavior over time, making the mean a reliable predictor, whereas seasonal cells are subject to gradual trends and periodic fluctuations, meaning recent observations carry more predictive information than the historical average.

It should be noted that MSE values are not directly comparable across dataset splits, as seasonal and non-seasonal cells differ in their underlying traffic variance. Comparisons are therefore made within each group independently.

5.2 ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a widely used statistical model for forecasting time series data. It is designed to capture patterns such as autocorrelation and trends in the data.

The model is defined by three parameters: p , d , and q .

- p : the number of autoregressive (AR) terms, which model the relationship between the current value and its past values.
- d : the degree of differencing used to make the series stationary.
- q : the number of moving average (MA) terms, which model the relationship between the current value and past forecast errors.

A seasonal extension (P, D, Q, m) can also be included, where m represents the number of time steps in one seasonal cycle. This allows the model to capture repeating patterns that occur every m periods.

The optimal parameters are selected by minimizing the Akaike Information Criterion (AIC), which balances model fit and complexity.

SARIMAX

SARIMAX is an extension of the ARIMA model. The S represents the seasonal component introduced previously, while the X indicates the inclusion of exogenous input features. Unlike ARIMA, which only models the historical behavior of the time series itself, SARIMAX can additionally incorporate external variables that may influence the forecasting.

Model Optimization

The following method was used to validate the ARIMA forecasting approach on the dataset. As described earlier, the data consists of two groups seasonal and non-seasonal. For each individual cell time series, the auto ARIMA function [Put python documentation link here] was applied to determine the parameter set that minimizes the AIC. The most frequently selected order across all cells was then chosen as the representative seasonal and non-seasonal configuration.

The proposed evaluation procedure consisted of three stages:

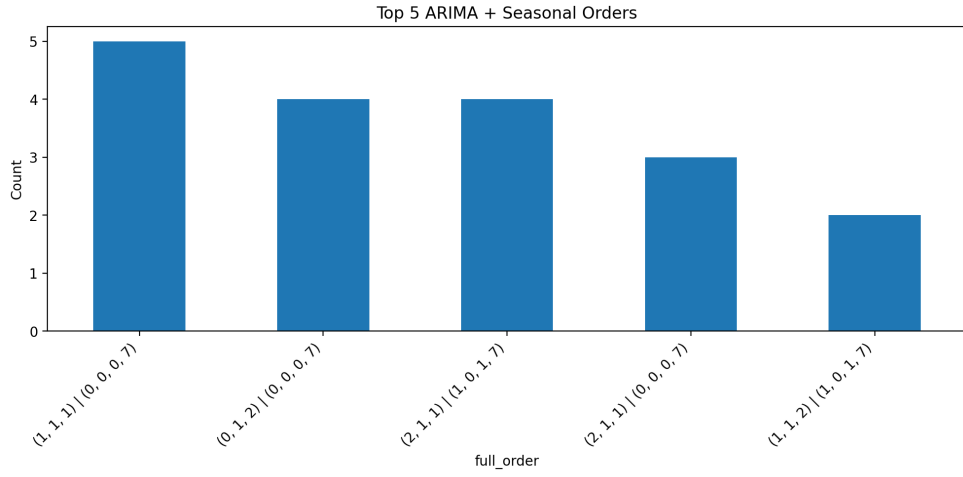
1. First, the baseline ARIMA configuration was determined using auto-ARIMA without any exogenous variables.
2. Second, forward feature selection was applied to the exogenous variables using the fixed baseline ARIMA order.
3. Finally, after selecting the optimal feature subset, the ARIMA coefficients and model parameters were re-estimated using the selected exogenous variables to obtain the final SARIMAX model.

baseline ARIMA configuration

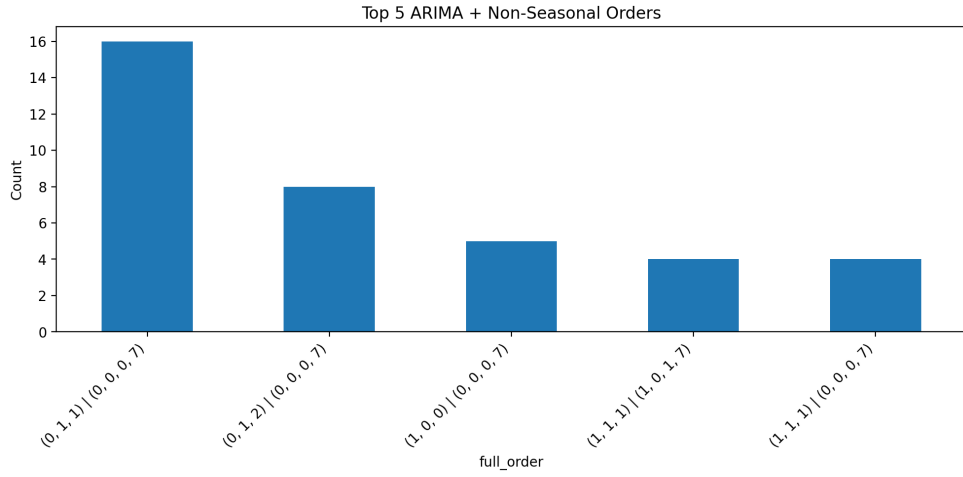
Figure 13 shows the frequency of the ARIMA orders selected by the auto-ARIMA procedure across all seasonal cell time series.

For the seasonal setting, the most frequently selected configuration was $\text{ARIMA}(1, 1, 1)(0, 0, 0, 7)$. This indicates that first-order differencing was consistently required, while the dominant structure was captured through short-term autoregressive and moving-average components.

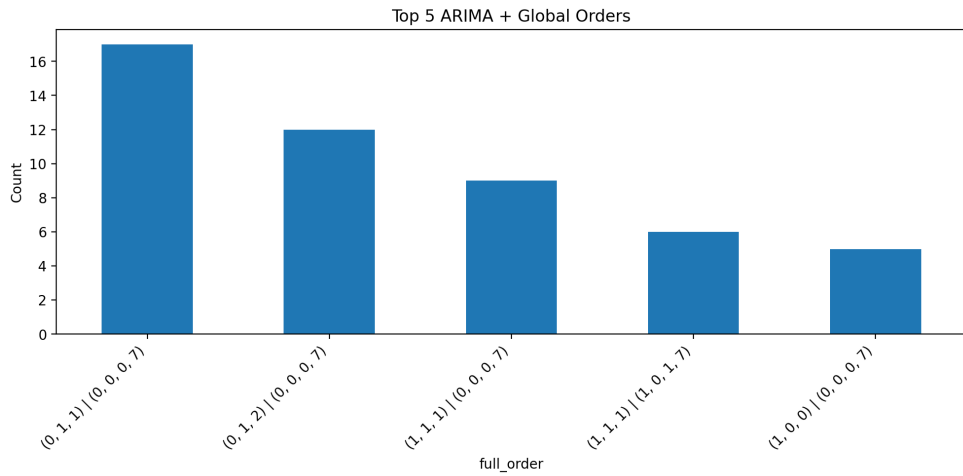
For the non-seasonal setting, the dominant configuration was $\text{ARIMA}(0, 1, 1)(0, 0, 0, 7)$. Compared to the seasonal case, the autoregressive component disappeared, suggesting weaker temporal dependence in the non-seasonal cells. However, first-order differencing,



(a) ARIMA order frequency for seasonal cell
S



(b) ARIMA order frequency for non-seasonal cell



(c) ARIMA order frequency for all cells

Figure 13: Arima order before feature selection

the moving-average and seasonal components remained consistently selected across both groups.

Feature selection

To evaluate the impact of exogenous variables on forecasting performance, a forward feature selection approach was used. The goal was to iteratively build a feature set that minimizes the mean squared error (MSE) across all time series (cells).

Table 5 shows the set of candidate features evaluated during the selection process.

Table 5: Candidate features evaluated during feature selection

Feature	Description
month	Month of the year
dayofweek	Day of the week
is_weekend	Weekend indicator variable
holiday	Public holiday indicator
lag1	Previous day traffic value
lag7	Traffic value from 7 days earlier
roll_mean_7	Rolling 7-day mean
roll_mean_14	Rolling 14-day mean
roll_std_7	Rolling 7-day standard deviation
roll_std_14	Rolling 14-day standard deviation
roll_min_7	Rolling 7-day minimum
roll_max_7	Rolling 7-day maximum
lag_diff_1_7	Difference between lag-1 and lag-7

To evaluate the impact of exogenous variables on forecasting performance, a forward feature selection approach is applied. The method starts with an empty feature set and iteratively adds the feature that yields the largest reduction in average test mean squared error (MSE) across all time series. Candidate features include calendar variables, lag-based features, rolling statistics, and differenced features. The process continues until no further improvement is observed, as shown in Algorithm 1.

Algorithm 1 Forward Feature Selection for ARIMA with Exogenous features

```
1: Initialize selected feature set  $S \leftarrow \emptyset$ 
2: Initialize remaining features  $R \leftarrow \{f_1, f_2, \dots, f_n\}$ 
3: Set best error  $E_{best} \leftarrow \infty$ 
4: while  $R \neq \emptyset$  do
5:    $f^* \leftarrow \emptyset$ 
6:    $E^* \leftarrow E_{best}$ 
7:   for each feature  $f \in R$  do
8:     Train ARIMA models with feature set  $S \cup \{f\}$  across all series
9:     Evaluate mean test error  $E_f$ 
10:    if  $E_f < E^*$  then
11:       $E^* \leftarrow E_f$ 
12:       $f^* \leftarrow f$ 
13:    end if
14:  end for
15:  if  $f^* \neq \emptyset$  then
16:    Add  $f^*$  to  $S$ 
17:    Remove  $f^*$  from  $R$ 
18:    Update  $E_{best} \leftarrow E^*$ 
19:  else
20:    Stop
21:  end if
22: end while
```

The forward selection procedure is applied separately to three model configurations: seasonal cells, non-seasonal cells, and the full dataset. Each configuration follows the same evaluation protocol, where features are added sequentially based on their impact on validation RMSE. The final selected feature set for each configuration is determined when no additional feature yields further improvement.

Table 6: Feature selection results across seasonal, non-seasonal, and all-cell models

Model Type	Selected Features (final step)	RMSE
Seasonal cells	roll_mean_7, roll_max_7	1.66
Non-seasonal cells	roll_min_7, lag_1, lag_diff_1_7	1.54
All cells	roll_min_7, lag_1, lag_diff_1_7	1.54

Feature selection results show strong overlap between the non-seasonal and all-cell configurations, which converge to the same optimal feature set. In contrast, seasonal cells rely more on short-term aggregated statistics such as rolling means and maxima, while non-seasonal cells benefit from lag-based and difference features that capture local temporal changes.

Across all configurations, performance improvements are primarily driven by short-term dependency features (lag-1 and lag-difference), suggesting that recent history dominates predictive power. Seasonal structure contributes less to overall error reduction, indicating

weak or inconsistent periodicity across most cells.

Post Feature Selection Orders

The resulting performance was:

(1, 1, 1) (0, 0, 0, 7) -j 1.6286740698148208 (0, 1, 2) (0, 0, 0, 7) -j 1.6421207574332402
(2, 1, 1) (1, 0, 1, 7) -j 1.8821604312943678 (2, 1, 1) (0, 0, 0, 7) -j 1.836335315622857 (1,
1, 2) (1, 0, 1, 7) -j 1.7244589444923621

Based on these results, the best-performing configuration for seasonal cells was:

$$(1, 0, 0)(0, 0, 0, 7)$$

with a final RMSE of 1.5291.

Non-seasonal cells

(0, 1, 1) (0, 0, 0, 7) -j 1.5406868111115992 (1, 0, 0) (0, 0, 0, 7) -j 1.499995505912129 (0,
0, 0) (0, 0, 1, 7) -j 1.5238978829181384 (0, 0, 2) (0, 0, 0, 7) -j 1.6096546642404328 (1, 0,
0) (1, 0, 0, 7) -j 1.5356852778076302

5.3 comparison

5.4 Ordinary Least Squares

Ordinary Least Squares (OLS) is a linear regression method that estimates parameters by minimizing the sum of squared errors between observed values and predictions.

In the simplest case with one feature, OLS fits a line of the form:

$$y = \beta_0 + \beta_1 x$$

where β_0 is the intercept and β_1 is the slope. These are chosen to minimize:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Unlike ARIMA, which models temporal dependence using past values of the time series, OLS does not assume any time structure. Each observation is treated as independent.

With multiple features, OLS generalizes to:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

or in matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Here, $\mathbf{X}$ is the feature matrix, $\boldsymbol{\beta}$ is the vector of learned coefficients, and $\boldsymbol{\epsilon}$ is the error term.

Each feature contributes additively to the prediction through its own weight, along with a single shared intercept. This allows OLS to incorporate engineered inputs such as lagged values, rolling statistics, or external variables, but it does not explicitly model time dynamics.

Feature selection

The same forward feature selection procedure used for the SARIMAX models (Algorithm 1) was applied to the OLS model. The candidate feature pool, evaluation metric, and stopping criterion were kept identical to ensure a fair comparison between the two approaches.

At each iteration, an OLS model was fitted using the current selected feature set and one additional candidate feature. The feature producing the largest reduction in validation RMSE was retained. The process continued until no further improvement in performance was observed.

Table 7: Feature selection results across seasonal, non-seasonal, and all-cell models

Model Type	Selected Features (final step)	RMSE
Seasonal cells	roll_mean_7, lag1, roll_mean_14, roll_std_14, holiday	1.45
Non-seasonal cells	roll_mean_14, lag1, roll_std_14, is_weekend	1.60
All cells	roll_min_7, lag_1, lag_diff_1_7	1.54

results

The results for the 3 data groups were as follows

5.5 XGboost

6 Results and Discussion

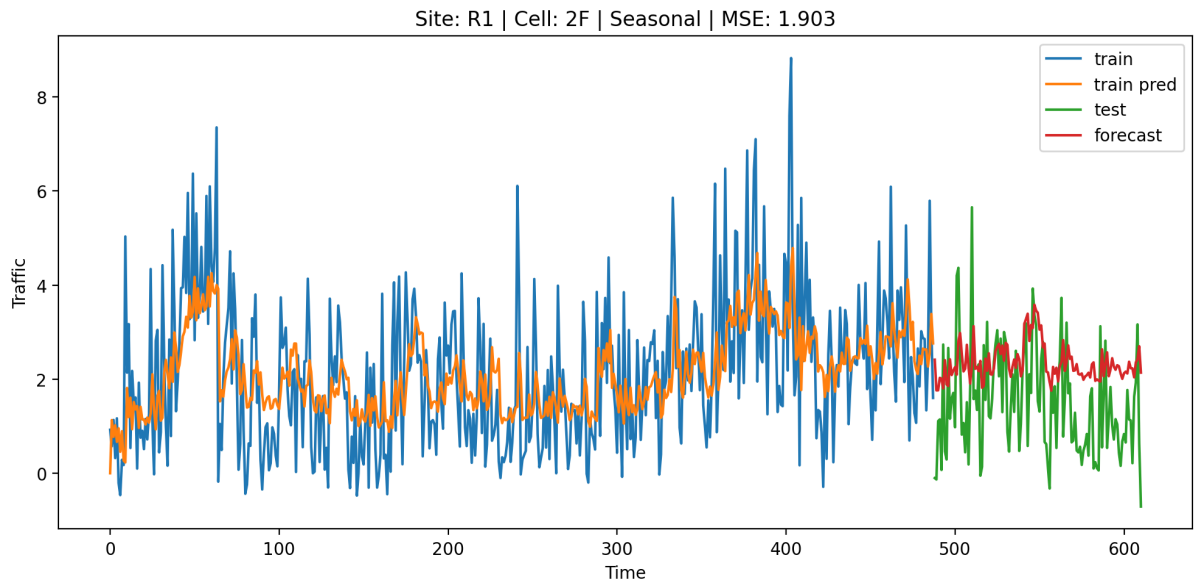
7 Conclusion

Summarize your findings.

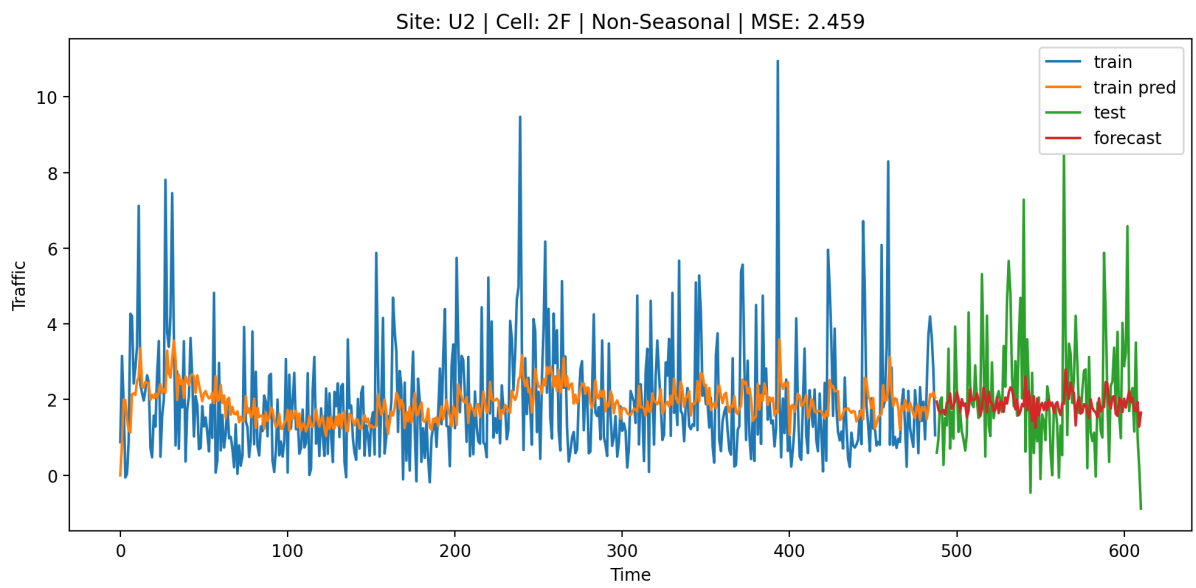
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(a) ARIMA forecasting seasonal cell



(b) ARIMA forecasting non-seasonal cell

Figure 14: Average data volume across cells for two towers